

No. of Functional Features Included in the Solution

Date	01 July 2026
Team ID	XXXX
Project Name	Rising Waters
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the Rising Waters flood prediction solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done/In Progress/Pending)	Marks Contribution
1	Data Preprocessing	Handles missing values, outliers, encoding, and feature scaling on rainfall and weather data.	Data Preprocessing Module	Done	0.75
2	Exploratory Data Analysis	Univariate/multivariate analysis, distribution plots, box plots, and heat maps to study feature patterns.	EDA / Visualization Module	Done	0.5
3	Model Training & Comparison	Trains and evaluates Decision Tree, Random Forest, KNN, and XGBoost classifiers.	Machine Learning Layer	Done	1.0
4	Best Model Selection	Selects XGBoost as best-performing model (96.55% accuracy) and saves it as floods.save.	Machine Learning Layer	Done	0.5
5	Home Page	Landing page introducing the flood prediction system to users.	Presentation Layer (Flask)	Done	0.25
6	Prediction Input Form	Collects rainfall, cloud visibility, and seasonal rainfall inputs with form validation.	Application Layer (Flask)	Done	0.5
7	Flood Risk Result Page	Displays Flood Chance / No Flood Chance result based on model prediction.	Application Layer (Flask)	Done	0.75
8	Cloud Deployment Readiness	Prepares the trained model and Flask app for scalable deployment on IBM Cloud.	Deployment Layer	In Progress	0.25

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Home Page, Prediction Input Form
2	Backend / Logic	3	Data Preprocessing, Model Training & Comparison, Best Model Selection
3	Database / Storage	1	Trained model file storage (floods.save)
4	API / Integration	1	Flask routing between input and result pages
5	Security / Authentication	0	Not yet implemented